

Deflection Measurement System in Flexible Bars Using Eulerian Video Magnification (EVM) Method

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ABSTRACT

Deflection testing plays a pivotal role in assessing a material's elasticity, which directly influences the safety and operational reliability of engineering structures. This study develops and evaluates a deflection measurement system employing a flexible bar test apparatus, integrating the Eulerian Video Magnification (EVM) technique. EVM enhances imperceptible displacements and motion changes in recorded video, enabling precise deflection observation. Experimental results indicate that EVM-based measurements recorded a peak deflection of 4.6693 mm and a minimum of -4.78208 mm, while accelerometer measurements yielded a peak of 3.7646 mm and a minimum of -3.7685 mm at 3,600 RPM. The results confirm that deflection magnitude increases proportionally with motor speed, aligning with prior research findings. These outcomes demonstrate the viability of EVM as a non-contact, effective method for detecting small-scale motion variations in materials that are difficult to monitor using conventional approaches. The proposed system improves the accuracy and efficiency of deflection measurements, offering significant implications for structural design and engineering analysis.

ABSTRAK

Pengujian defleksi merupakan tahap penting dalam menilai elastisitas suatu material, yang secara langsung memengaruhi keamanan dan keandalan operasional struktur teknik. Penelitian ini mengembangkan dan menguji sistem pengukuran defleksi menggunakan perangkat uji batang lentur yang menerapkan teknik Eulerian Video Magnification (EVM). EVM memperkuat perpindahan dan perubahan gerakan yang tidak terlihat pada rekaman video, sehingga memungkinkan pengamatan defleksi dengan tingkat ketelitian tinggi. Hasil eksperimen menunjukkan bahwa pengukuran berbasis EVM mencatat defleksi puncak sebesar 4,6693 mm dan nilai minimum -4,78208 mm, sedangkan pengukuran menggunakan akselerometer menghasilkan defleksi puncak 3,7646 mm dan nilai minimum -3,7685 mm pada kecepatan 3.600 RPM. Temuan ini menegaskan bahwa besarnya defleksi meningkat seiring bertambahnya kecepatan putaran motor, sejalan dengan temuan penelitian sebelumnya. Hasil tersebut menunjukkan bahwa EVM layak digunakan sebagai metode non-kontak yang efektif untuk mendeteksi variasi gerakan berskala kecil pada material yang sulit diamati melalui metode konvensional. Sistem ini berpotensi meningkatkan akurasi dan efisiensi pengukuran defleksi, dengan implikasi signifikan bagi perancangan dan analisis struktur teknik.

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1. Introduction

Deflection testing is a critical procedure for evaluating the elasticity of materials, which directly affects the safety, stability, and performance of engineering structures. Deflection refers to the degree to which a material or structure deforms under a load. In practice, the procedure involves applying a load to the material or structure and measuring the resulting deformation. The importance of deflection testing lies in its ability to provide insight into a material's behavior under applied loads. Understanding this behavior allows

engineers and designers to determine whether a material meets the required elasticity criteria for specific applications. For example, in mechanical design, an optimal level of elasticity is essential to ensure that machines operate safely and efficiently without excessive deformation. Deflection testing also plays a key role in designing structures capable of withstanding intended loads without compromising performance. By quantifying deflection, engineers can make informed decisions to optimize structural safety and operational reliability (Ahmad, 2006). Furthermore, deflection is often influenced by vibration, which is commonly present in moving machinery. Even low-intensity vibrations can accumulate over time, potentially causing structural damage or reducing mechanical performance (Melchers, 2003). For this reason, monitoring vibration is essential to identify potential hazards and implement preventive measures. In many structures—whether in buildings or operating machinery—small-scale vibrations are often imperceptible to the naked eye. If left unmonitored, such vibrations can lead to progressive deflection and eventual failure. To address this challenge, the Eulerian Video Magnification (EVM) method offers a promising solution. In recent years, various approaches have been developed to detect subtle motion within video sequences. Such micro-movements, invisible to human observation, require specialized analysis techniques to be measured accurately. The primary objective of motion analysis in this context is to track temporal changes in image vectors over time. EVM, as described by Wadhwa et al. (2017), is a computational algorithm designed to detect and analyze subtle motion and minute variations in video sequences. Functioning as a “digital microscope,” it operates without optical magnification, instead using software to process standard video input. The algorithm amplifies near-invisible temporal changes, enabling detailed observation of micro-motions and subtle color variations. This capability has broad applications in healthcare, biology, mechanical engineering, and civil engineering. In vibration monitoring, EVM can be combined with a flexible bar testing apparatus to measure structural response under controlled excitation from a rotating motor. The present study applies this approach to evaluate the feasibility of EVM as a non-contact alternative to conventional deflection measurement methods.

2. Research Methodology

2.1 Research Workflow

The experimental procedure is summarized in Figure 1. The process begins with the selection of primary components and the test specimen, followed by configuring the experimental setup, instrument calibration, and adjusting motor speed variations. Deflection measurements are then conducted using two approaches: an indirect method based on Eulerian Video Magnification (EVM) and a direct method using an accelerometer. The collected data are subsequently analyzed, and conclusions are drawn based on the comparative evaluation of both measurement techniques.

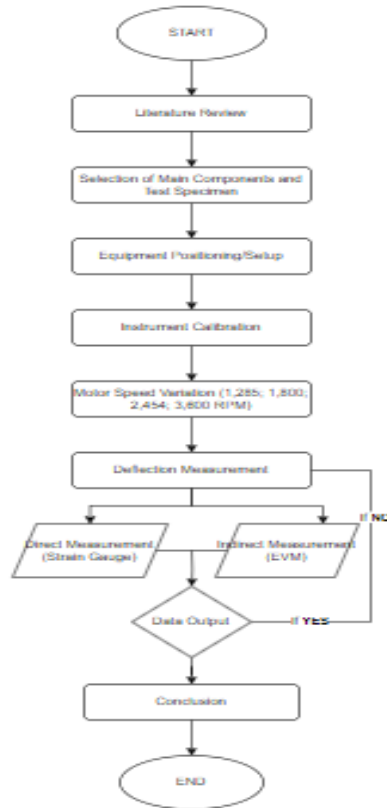


Figure 1. Research Workflow Diagram

2.2 Selection of Main Components and Test Specimen

The selection of main components and the test specimen is a critical step in experimental and engineering research, particularly in the context of system design and testing. Appropriate component selection ensures that the experimental setup meets the intended operational requirements, supports accurate measurement, and provides reliable data for subsequent analysis.

2.2.1 Vibration Generation System Design

The vibration generation system is designed using a DC electric motor with an adjustable rotational speed ranging from 120 to 1,800 RPM. An eccentric mass is mounted on the motor shaft to produce a centrifugal force, which transmits vibration to the flexible bar. A regulated power supply is employed to control motor speed, enabling observation of vibration frequency variations.

2.2.2 Design of the Flexible Bar Test Apparatus

Following the design of the vibration generation system, the next stage involves constructing the flexible bar test apparatus. This design stage includes determining the apparatus dimensions, selecting an appropriate frame type, and defining the support configuration for the flexible bar. The selected support arrangement consists of one fixed clamp support and one roller support. The clamp support provides rigidity at one end of the bar, while the roller support allows horizontal translation at the opposite end, enabling accommodation of axial loads. Table 1. Specifications of the Flexible Bar Test Apparatus

Component	Dimensions	Material	Description
Main Frame	100 × 50 × 40 cm	Wood	Provides structural rigidity to withstand vibration from the motor and applied loads
Adjustable Stand	Length 30–100 cm	Wood	Adjustable to accommodate various test specimen sizes
Flexible Bar	Variable (0.5–10 kg)	Steel/Iron	Test specimen for deflection observation
Motor Assembly	10 mm shaft diameter	Steel + electronics	Equipped with eccentric mass for vibration generation
Roller Support	50 × 30 cm	Steel	Allows horizontal translation at the free end to accommodate axial forces
Clamp Support	20 × 15 × 10 cm	Steel	Provides rigidity and supports bending moments at one end

2.2.3 Deflection Measurement System

Deflection measurements of the flexible bar are conducted using two methods: direct measurement with an accelerometer and indirect measurement using the Eulerian Video Magnification (EVM) technique.

A. Accelerometer

An accelerometer is a sensor that measures acceleration in an object. In the context of deflection testing, the accelerometer detects changes in the position and velocity of the flexible bar. The recorded acceleration data are then processed to derive deflection values.

B. Eulerian Video Magnification (EVM)

EVM is a video-processing method that magnifies subtle positional changes that are otherwise invisible to the naked eye. In deflection testing, this method is employed to visually detect and measure the positional changes of the flexible bar with enhanced clarity.

Tabel 2. Perbandingan metode pengukuran akselerometer dan EVM

Aspect	Direct Measurement (Accelerometer)	Indirect Measurement (EVM)
Operating Principle	Measures acceleration and converts it into deflection data	Analyzes video to detect small positional changes
Contact Requirement	Requires direct sensor attachment	Non-contact
Response Speed	Real-time	Requires post-processing
Equipment	Accelerometer, DAQ, analysis software	Camera, EVM software, computer
Measurement Accuracy	High, but dependent on calibration and sensor placement	High, but dependent on camera resolution and video quality
Ease of Use	Requires careful calibration	Relatively simple setup
Suitable Applications	High-frequency dynamic testing	Small or slow-motion measurements
Main Advantage	Provides fast, accurate data for dynamic testing	Non-intrusive, does not affect mechanical

		properties
Main Limitation	May alter bar properties due to added mass	Requires longer data processing time

2.2.4 Equipment Positioning and Setup

The primary purpose of the equipment setup is to arrange each component in an optimal position to ensure:

1. Testing conditions are consistent with the experimental design.
2. Deflection measurements can be performed with high accuracy.
3. Stability and safety are maintained throughout the testing process.
4. External interference—such as environmental vibrations or unintended equipment movement—is minimized.

The main components to be positioned in the setup include:

1. Flexible Bar – Positioned horizontally between the supports.
2. Clamp Support – Installed at one end of the apparatus to hold the bar rigidly in place and ensure stability.
3. Roller Support – Installed at the opposite end, allowing horizontal translation to accommodate axial movement.
4. Motor (Vibration Source) – Positioned at the midpoint of the flexible bar to achieve a balanced and precise vibration effect.
5. Camera – Mounted in a fixed position to capture the entire length of the bar. The distance from the apparatus is adjusted to record fine motion details without distortion.

Additional considerations during equipment setup include:

1. Environmental Conditions:
 - a. Avoid exposure to external vibrations from surrounding equipment or structures.
 - b. Ensure adequate lighting for optimal video capture quality.
2. Safety and Stability:
 - a. Secure all components firmly to prevent damage or disruption during testing.

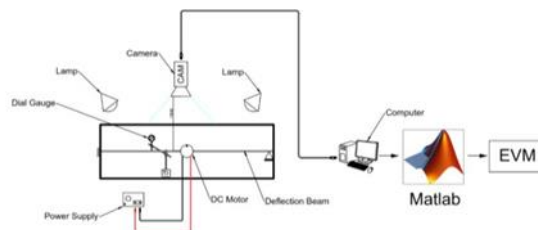


Figure 2. Experimental Schematic

Table 3. Equipment Positioning and Setup

Component	Position in the Test Apparatus	Function	Special Notes
Flexible Bar	Between clamp support and roller support	Primary object for deflection measurement	Ensure the bar is perfectly horizontal
Clamp Support	Fixed end of the flexible bar	Holds the bar rigidly and stably	Secure firmly to prevent movement
Roller Support	Free end of the flexible bar	Supports the bar and allows horizontal	Ensure smooth and flexible roller movement

		translation	
Vibration Source/Load	Above the bar (midpoint or bar end)	Generates load or vibration on the bar	Position according to experimental requirements
Accelerometer	Midpoint of the bar (or critical deflection point)	Measures acceleration due to vibration/load	Ensure the sensor is properly calibrate
Camera (EVM)	Parallel to the bar at an appropriate distance	Records bar movement for visual analysis	Use a tripod to ensure camera stability

2.2.5 Instrument Calibration

The next stage involves calibrating the apparatus to ensure that the testing equipment produces accurate, consistent, and standard-compliant data. Calibration is also necessary to adjust for or correct systematic errors in the equipment, such as sensor reading inaccuracies or discrepancies in measurement scale. The components requiring calibration include:

1. Accelerometer – Calibrate sensitivity and offset to ensure precise acceleration measurements.
2. Camera – Adjust focus, frame rate, and alignment to achieve optimal image clarity and measurement accuracy.
3. Vibration/Load Source – Verify rotational speed and load output to maintain consistent excitation conditions.
4. Clamp and Roller Supports – Ensure alignment and rigidity to prevent unwanted movement that could affect measurement accuracy.

2.3 Motor Speed Variation

Motor speed variation is employed in the experiment to create diverse testing conditions. The primary objective is to generate vibrations or dynamic loads with different frequencies in order to evaluate the response of the flexible bar. By varying motor speeds, it is also possible to analyze the relationship between motor rotation speed, vibration frequency, and the resulting deflection of the flexible bar. The motor speed parameter is expressed in revolutions per minute (RPM).

2.4 Deflection Testing Procedure

Once all components of the testing apparatus have been correctly assembled—including the flexible bar, clamp and roller supports, and measurement sensors—the deflection testing can be conducted to determine the extent of deformation in the bar under applied vibration. This testing stage also serves to validate the performance of both measurement systems: the direct method using an accelerometer and the indirect method using Eulerian Video Magnification (EVM). The final outcome of this procedure is to assess whether camera-based measurement using the EVM method can serve as a reliable alternative to conventional deflection measurement techniques.

3. Results and Discussion

3.1 Design and Development of the Deflection Measurement System

Recent studies emphasize the importance of integrating both contact and non-contact measurement systems in structural deflection testing. According to Chen et al. (2020), non-contact video-based techniques such as EVM offer significant advantages in terms of preserving the mechanical integrity of the specimen while enabling high-resolution displacement tracking. Similarly, Siringoringo and Fujino (2006) demonstrated that accelerometer-based measurements provide high temporal accuracy, making them

suitable for validating non-contact methods. The combination of these approaches ensures both reliability and robustness in deflection monitoring systems, especially in applications involving small-scale vibrations.

3.1.1 System and Apparatus Design

The deflection measurement system consists of a flexible bar testing apparatus with clamp and roller supports. The vibration source is a DC motor, while the measurement system comprises an accelerometer for direct measurement and a camera for indirect measurement using Eulerian Video Magnification (EVM).

A. System Design Diagram



Figure 3. Design of the Deflection Measurement System

B. Components Used in the Design

1. Flexible Bar – Made of low-carbon steel.
2. Supports (Clamp and Roller) – The clamp support rigidly holds one side of the bar, while the roller support allows free movement on the opposite side.
3. Vibration Source – A DC motor with a speed range of 1,285–3,600 RPM, equipped with an eccentric mass to generate vibration.
4. Measurement System – Accelerometer and camera (EVM).
5. Control System – The DC motor is controlled via a power supply, and the accelerometer is operated using an Arduino microcontroller.
6. Software – MATLAB is used to process video data from the camera, while Python programming is applied to acquire and process accelerometer data.

C. Apparatus Illustration



Figure 4. Flexible Bar

D. Design Workflow

The design process involves the following steps:

1. Needs Analysis – Identifying the parameter to be measured, namely deflection of the flexible bar.
2. Component Selection.
3. System Design.
4. Assembly/Construction.
5. Preliminary Testing.

Table 4. Component Specifications

Component	Specification	Function	Notes
Flexible Bar	Low-carbon steel, 500 × 20 × 5 mm	Test medium for deflection measurement	Controlled via the supports
Clamp Support	Steel material, strong clamp	Holds the bar rigidly	Placed at one end of the bar
Roller Support	Steel material, free rotation	Allows free movement of the bar	Placed at the opposite end of the bar
DC Motor	1,000–5,000 RPM, 12V	Vibration/dynamic load source	Equipped with an eccentric mass
Accelerometer	MPU6050, ±16g	Measures acceleration/deflection directly	Positioned at the midpoint of the bar
Camera (EVM)	1080p, 60 fps	Records video for indirect deflection analysis	Positioned parallel to the bar
Arduino Uno	Microcontroller	Controls the motor and reads sensor data	Connected to a computer

3.2 Testing of the EVM-Based Measurement System

Deflection testing of the flexible bar in this study was carried out using two different approaches: indirect measurement with Eulerian Video Magnification (EVM) and direct measurement with an accelerometer. For the EVM method, a camera was used to record positional changes or movements of the flexible bar resulting from applied loading. The recorded video was subsequently processed using MATLAB software to convert the visual data into numerical form. This processing step enabled the quantification of deflection magnitudes that occurred in the flexible bar. As a validation step to ensure the accuracy of the EVM-derived data, additional testing was performed using the direct measurement method with an accelerometer. The data obtained from the accelerometer were processed using the Python programming language, which was configured to operate the accelerometer and convert the measurements into numerical deflection data ready for analysis.

A. Testing at 3,600 RPM with Clamp Support

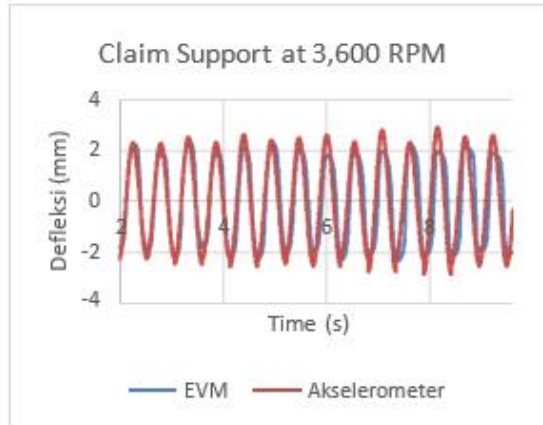


Figure 5. Vibration Signal Comparison at Clamp Support, 3,600 RPM

Figure 5 presents the comparison between EVM and accelerometer measurement results at the clamp support with a motor speed of 3,600 RPM. Using the EVM method, the recorded peak deflection was 2.3232 mm, with the lowest deflection measured at -2.3551 mm. In comparison, the accelerometer recorded a peak deflection of 2.8687 mm and a minimum deflection of -2.8658 mm.

B. Testing at 3,600 RPM with Roller Support

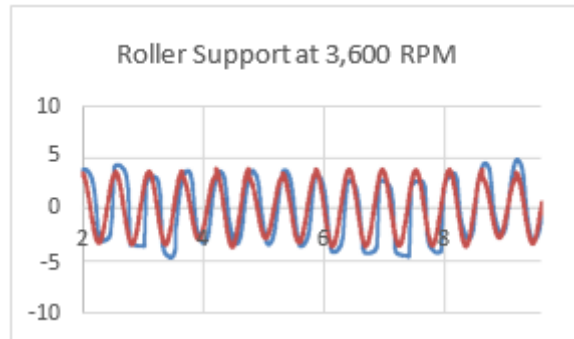


Figure 6. Vibration Signal Comparison at Roller Support, 3,600 RPM

Figure 6 presents the comparison between EVM and accelerometer measurements at the roller support with a motor speed of 3,600 RPM. Using the EVM method, the recorded peak deflection was 4.6693 mm, with the lowest deflection measured at -4.78208 mm. In comparison, the accelerometer recorded a peak deflection of 3.7646 mm and a minimum deflection of -3.7685 mm.

3.3 Discussion of Design and Measurement Results

The deflection measurement results obtained using Eulerian Video Magnification (EVM) and direct accelerometer measurement indicate a clear trend: as motor rotational speed increases, the deflection amplitude of the flexible bar also increases. This relationship is consistent with the findings of Basori et al. (2015), who conducted similar research and observed that higher excitation frequencies produce greater deflection magnitudes in flexible structures. The differences in the absolute deflection values recorded by the two methods can be attributed to the inherent characteristics of each approach. Accelerometer-based measurement offers high temporal accuracy and direct contact sensing but may slightly influence the system's dynamic behavior due to the added sensor mass. In contrast, the EVM method avoids physical interference, making it ideal for non-contact monitoring, but its accuracy depends heavily on video resolution, camera stability, and processing parameters in MATLAB.

4. Conclusions

The design and development of the deflection measurement system were successfully completed. Testing results demonstrated that the greatest deflection occurred at the highest motor rotational speed of 3,600 RPM, confirming that deflection amplitude increases proportionally with motor speed. The comparison between the Eulerian Video Magnification (EVM) method and direct accelerometer measurement showed that EVM can serve as a viable non-contact alternative to conventional deflection measurement methods. While the accelerometer provided precise, real-time measurements, EVM offered the advantage of not requiring physical contact with the specimen, making it suitable for applications where direct sensor installation is impractical or could interfere with the system's behavior. The integration of both methods in this study strengthens the reliability of deflection measurements, providing a framework for accurate, efficient, and flexible structural monitoring in engineering applications.

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